1. Clinical Question & Core Hypothesis

Clinical Question

The Ontologist highlights that PARP inhibition (Olaparib) intersects tumor immune context and genomic classifiers, yet TLS- or TIL-rich microenvironments do not always produce durable antitumor responses or clinical benefit. The key puzzle is: why does an apparently immune-receptive TME (TLS/TIL-rich) sometimes fail to translate PARP-induced DNA damage into effective antitumor immunity and measurable therapy benefit?

Working Hypothesis

Among TLS-like lymphoid aggregate–rich prostate tumors, high Decipher GC scores identify intrinsic transcriptional programs that impair neoantigen presentation and TLS function, blunting antitumor immunity elicited by PARP-inhibitor–induced DNA damage. Consequently, TLS-high tumors with high Decipher GC will exhibit poorer therapy_response to Olaparib than TLS-high/Decipher-low tumors, indicating a need for combinatorial immunomodulatory strategies.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	Decipher GC, Olaparib
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

Decipher GC: Decipher GC" path because a genomic-risk classifier can mechanistically mark intrinsic tumor programs (proliferation, immune suppression) that modulate immune engagement.

Olaparib: Olaparib with immune-context modulation; therapy_response is the most direct clinical consequence to interrogate mechanistic failure of immune engagement after PARP-induced DNA damage.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	CENPA	Supporting modifier from knowledge graph
Secondary	EDIP (insulin/metabolic response markers)	Supporting modifier from knowledge graph

Secondary	f/tPSA	Supporting modifier from knowledge graph
Secondary	urinary biomarkers	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: , and Decipher GC–associated signatures; (c) in vitro organoid–immune co-culture assays assessing T cell activation/killing under Olaparib with or without immunomodulators; (d) in vivo models engineered to form TLS-like niches to test combination therapies (PARP inhibitor + immune checkpoint blockade or TLS-promoting agents); (e) candidate biomarker readouts (MHC-I, CXCL13, CD8+ T cell density, urinary and plasma inflammatory/metabolic markers) to monitor response and stratify patients.

Prediction 2: At the tissue level, tumors with the imaging feature but low anchor gene activity will show immune cells confined to structural peripheries rather than infiltrating tumor nests.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- CENPA: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- EDIP (insulin/metabolic response markers): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.